

Recursively differentiable quasigroups of orders 14 and 18: the last open cases of the Couselo–González–Markov–Nechaev conjecture

Alejandro Lizardi*

June 2026

Periapsis 

Abstract

A quasigroup $(\Omega, \cdot)$ is *recursively differentiable* if its first recursive derivative $x * y := y \cdot (x \cdot y)$ is again a quasigroup. In 1998 Couselo, González, Markov, and Nechaev conjectured that a recursively differentiable quasigroup of order q exists for every $q \notin \{2, 6\}$, and verified this for all q except $q \in \{14, 18, 26, 42\}$. The case $q = 42$ was settled in 2008 and the case $q = 26$ in 2026, leaving $q = 14$ and $q = 18$ as the only open orders. We settle both by explicit construction. We exhibit a recursively differentiable quasigroup of order 14 and one of order 18, each invariant under a fixed-point-free automorphism with two regular orbits (a pair of 7-cycles and a pair of 9-cycles, respectively). Both are found by a prescribed-automorphism search and certified by the $O(n^2)$ test that defines the property; the full Cayley tables and their recursive derivatives are printed in the paper and re-checked by a standalone verifier. Together with the known cases this completes the conjecture: a recursively differentiable quasigroup of order q exists for every $q \notin \{2, 6\}$, and for no other order is the question open. The two orders resisted the standard perfect-Mendelsohn-design constructions for an arithmetic reason we make explicit (for those design constructions, $14, 18 \not\equiv 0, 1 \pmod{4}$ and $\not\equiv 0, 1 \pmod{5}$ — a gap in the *known methods*, not an obstruction to existence), which is why an ad-hoc, computer-assisted construction is the natural route to them.

2020 Mathematics Subject Classification. 20N05 (loops, quasigroups); 05B15 (orthogonal arrays, Latin squares); 94B25 (combinatorial codes); 05B30 (other designs, configurations).

Keywords. recursively differentiable quasigroup; recursive derivative; quasigroup core; recursive MDS code; Latin square; prescribed automorphism; perfect Mendelsohn design.

1 Introduction

A *quasigroup* of order n is a finite set Ω with a binary operation $\cdot$ whose Cayley table is a Latin square: every row and every column is a permutation of Ω . Equivalently, both equations $a \cdot x = b$ and $y \cdot a = b$ have unique solutions for all a, b .

*Independent researcher, alej@lizardi05@gmail.com. The constructions and computations in this paper were AI-assisted by Claude, under the Periapsis research structure. Both main results are explicit finite objects whose defining property is verifiable in $O(n^2)$ arithmetic; a self-contained verifier is provided with the ancillary files (Section 7).

Given a quasigroup $(\Omega, \cdot)$, Couselo, González, Markov, and Nechaev [1] defined a sequence of *recursive derivatives* $(\Omega, *_n)$ by

$$a *_{-2} b = a, \quad a *_{-1} b = b, \quad a *_n b = (a *_n b) \cdot (a *_n b) \quad (n \geq 0). \quad (1)$$

Thus $a *_0 b = a \cdot b$ is the original operation, and the *first recursive derivative* is

$$a *_1 b = (a *_{-1} b) \cdot (a *_0 b) = b \cdot (a \cdot b). \quad (2)$$

A quasigroup is *recursively n -differentiable* if $(\Omega, *_k)$ is a quasigroup for every $0 \leq k \leq n$; a recursively 1-differentiable quasigroup is called simply *recursively differentiable*. In the loop-and-quasigroup open-problems list [4] this object appears under the name *quasigroup with a quasigroup core*: the question, proposed by Syrbu at Loops '03, asks for which small orders a quasigroup Q exists such that the operation $x * y$ defined by (2) is itself a quasigroup operation. (The list writes the operation as $x * y = y - xy$, where the dash denotes the quasigroup multiplication, not subtraction; a general quasigroup has no inverses, so $\cdot$, not $^{-1}$, is the only operation available.)

By a theorem of [1], recursive $(n-3)$ -differentiability of a quasigroup of order q is equivalent to the existence of a full recursive $[n, 2, n-1]_q$ MDS code; the case $n = 4$, i.e. recursive differentiability, corresponds to a recursive $[4, 2, 3]_q$ code. Constructing an ordinary $[4, 2, 3]_q$ code amounts to a pair of orthogonal Latin squares of order q , which exist for all $q \notin \{2, 6\}$ (Bose–Shrikhande–Parker); the *recursive* constraint is what makes the differentiable version hard.

Conjecture 1.1 (Couselo–González–Markov–Nechaev [1]). *For every $q \notin \{2, 6\}$ there exists a recursively differentiable quasigroup of order q .*

The originators verified Conjecture 1.1 for all q except $q \in \{14, 18, 26, 42\}$. Markov, Nechaev, Skazhenik, and Tveritinov [2] settled $q = 42$ in 2008 with an explicit construction, and Klimov [3] settled $q = 26$ in 2026 via a perfect cyclic Mendelsohn design. This left exactly two open orders, $q = 14$ and $q = 18$. Our main result removes them.

Theorem 1.2. *There exists a recursively differentiable quasigroup of order 14 and a recursively differentiable quasigroup of order 18. Explicit Cayley tables are given in Sections 5 and 6.*

Corollary 1.3. *Conjecture 1.1 holds: a recursively differentiable quasigroup of order q exists for every $q \notin \{2, 6\}$, and these are exactly the orders for which one exists.*

Proof. The orders $q \in \{2, 6\}$ admit no pair of orthogonal Latin squares, hence no recursive $[4, 2, 3]_q$ code and no recursively differentiable quasigroup. All orders $q \notin \{2, 6, 14, 18, 26, 42\}$ are [1]; $q = 42$ is [2]; $q = 26$ is [3]; and $q \in \{14, 18\}$ is Theorem 1.2. $\square$

The verification underlying Theorem 1.2 is entirely elementary: each claimed object is a finite Cayley table, and the recursive-differentiability property is checked in $O(n^2)$ arithmetic directly from the definition (Section 2). The two tables were produced by a prescribed-automorphism backtracking search (Section 4); the search is how the objects were *found*, but it is not part of their *certification* — a reader may discard the search entirely and check the printed tables by hand or in a few lines of code.

Why 14 and 18 were the residue. Section 3 records the elementary arithmetic reason the standard constructions miss exactly these two orders, which both explains why they remained open and motivates the ad-hoc route we take.

2 The recursive derivative as a table operation

Throughout we identify a quasigroup of order n with its Cayley table $L: \Omega \times \Omega \rightarrow \Omega$, $\Omega = \{0, 1, \dots, n-1\}$, where $L[a][b] = a \cdot b$; L is a Latin square. From (2), the first recursive derivative has Cayley table

$$D[a][b] = a *_1 b = b \cdot (a \cdot b) = L[b][L[a][b]]. \quad (3)$$

Thus the derivative is obtained from L by a single nonlinear substitution: in row a , column b , look up the product $v = L[a][b]$, then read off $L[b][v]$. A quasigroup $(\Omega, \cdot)$ is recursively differentiable precisely when both L and the derived table D of (3) are Latin squares. This is the whole certificate, and it is checkable in $O(n^2)$ time.

Remark 2.1. The cyclic group $\mathbb{Z}_n$ has $L[a][b] = a + b$, so $D[a][b] = b + (a + b) = a + 2b$. As a map of b for fixed a this is a bijection iff $\gcd(2, n) = 1$, i.e. iff n is odd. Hence $\mathbb{Z}_n$ (and more generally any quasigroup that is “linear” over an abelian group with the doubling map appearing) is recursively differentiable exactly for odd n , and is *not* for even n . In particular the two orders we treat, being even, admit no such linear solution: a genuinely nonlinear table is required.

3 Why orders 14 and 18 resisted the standard constructions

The most productive known route to recursively differentiable quasigroups, used by Klimov [3] for $q = 26$, is the *directed standard construction* from a perfect cyclic Mendelsohn design. A $(v, k, 1)$ -PMD yields, by reading off consecutive triples of each cyclically ordered block, a recursively $(k - 3)$ -differentiable quasigroup of order v ; in particular a $(v, 4, 1)$ -PMD gives a recursively differentiable quasigroup of order v . The relevant existence facts are classical: a $(v, 4, 1)$ -PMD exists for all $v \equiv 0, 1 \pmod{4}$ (except $v \in \{4, 8\}$), and a $(v, 5, 1)$ -PMD for all $v \equiv 0, 1 \pmod{5}$ (except $v \in \{6, 10, 15, 20\}$); see the survey in [3] and the references therein.

This is exactly where 14 and 18 fall through:

$$14 \equiv 2 \pmod{4}, \quad 18 \equiv 2 \pmod{4}, \quad 14 \equiv 4 \pmod{5}, \quad 18 \equiv 3 \pmod{5},$$

so neither admits a $(v, 4, 1)$ - or a $(v, 5, 1)$ -PMD, and the block-size-4 and -5 PMD routes are simply unavailable. (By contrast $26 \equiv 1 \pmod{5}$, which is precisely what let [3] build a $(26, 5, 1)$ -PMD.) The other published families also miss the two orders: linear-recursive-sequence constructions live over prime-power alphabets and $14 = 2 \cdot 7$, $18 = 2 \cdot 3^2$ are not prime powers; the pseudogeometry method of [2] produced only $q = 42$. Klimov [3] accordingly lists 14 and 18 as the remaining open cases, to be approached “by other classes of methods.” The prescribed-automorphism search of the next section is such a method.

We stress that this is a statement about *known constructions*, not an obstruction to existence: most even orders *do* carry recursively differentiable quasigroups (e.g. 10, 12, 20, 22, 24, $\dots$ are all known), so there is no parity or congruence barrier in principle — 14 and 18 are merely the awkward composite orders that the congruence-restricted design constructions happen to skip. Theorem 1.2 confirms that existence holds for them too.

4 The construction method

Both objects were found by a backtracking search for a Latin square L such that the derived table D of (3) is also Latin, constrained to be invariant under a prescribed automorphism. A

permutation g of Ω is an *automorphism* of $(\Omega, \cdot)$ if $L[g(a)][g(b)] = g(L[a][b])$ for all a, b ; such a g partitions the n^2 cells of L into orbits under $(a, b) \mapsto (g(a), g(b))$, and fixing the value of one cell forces the values on its whole orbit. Searching only over orbit representatives shrinks the space by a factor of $\text{ord}(g)$ and tends to produce highly structured solutions.

We take g to be a fixed-point-free permutation with two regular orbits: for $n = 14$,

$$g_{14} = (0\ 1\ 2\ 3\ 4\ 5\ 6)(7\ 8\ 9\ 10\ 11\ 12\ 13),$$

two 7-cycles; for $n = 18$,

$$g_{18} = (0\ 1\ 2\ 3\ 4\ 5\ 6\ 7\ 8)(9\ 10\ 11\ 12\ 13\ 14\ 15\ 16\ 17),$$

two 9-cycles. (The choice mirrors the factorization $n = 2m$ with m the odd part: $14 = 2 \cdot 7$, $18 = 2 \cdot 9$.) The search assigns cells in a most-constrained-first order, propagates each assignment across its g -orbit and through the partial derivative table D (a conflict in either L or D prunes the branch immediately), and uses randomized restarts. The order-14 table was found in seconds; the order-18 table required randomized restarts and about half an hour of search on a laptop. Implementation, parameters, and run logs are in the ancillary files (Section 7); none of it is needed to verify the results.

5 A recursively differentiable quasigroup of order 14

Theorem 5.1. *The Cayley table L_{14} below is a quasigroup of order 14 whose first recursive derivative D_{14} is also a quasigroup. Hence a recursively differentiable quasigroup of order 14 exists.*

The base table L_{14} (rows indexed by the left factor $a = 0, \dots, 13$ top to bottom, columns by the right factor $b = 0, \dots, 13$ left to right):

0	2	7	5	10	12	9	6	4	13	3	1	11	8
10	1	3	8	6	11	13	9	0	5	7	4	2	12
7	11	2	4	9	0	12	13	10	1	6	8	5	3
13	8	12	3	5	10	1	4	7	11	2	0	9	6
2	7	9	13	4	6	11	0	5	8	12	3	1	10
12	3	8	10	7	5	0	11	1	6	9	13	4	2
1	13	4	9	11	8	6	3	12	2	0	10	7	5
8	4	13	0	2	7	5	10	3	9	1	12	6	11
6	9	5	7	1	3	8	12	11	4	10	2	13	0
9	0	10	6	8	2	4	1	13	12	5	11	3	7
5	10	1	11	0	9	3	8	2	7	13	6	12	4
4	6	11	2	12	1	10	5	9	3	8	7	0	13
11	5	0	12	3	13	2	7	6	10	4	9	8	1
3	12	6	1	13	4	7	2	8	0	11	5	10	9

Its first recursive derivative $D_{14}[a][b] = L_{14}[b][L_{14}[a][b]]$:

0	3	13	10	12	4	2	5	1	7	11	6	9	8
3	1	4	7	11	13	5	9	6	2	8	12	0	10
6	4	2	5	8	12	7	11	10	0	3	9	13	1
8	0	5	3	6	9	13	2	12	11	1	4	10	7
7	9	1	6	4	0	10	8	3	13	12	2	5	11
11	8	10	2	0	5	1	12	9	4	7	13	3	6
2	12	9	11	3	1	6	0	13	10	5	8	7	4
4	6	3	13	9	11	8	1	7	12	10	0	2	5
9	5	0	4	7	10	12	6	2	8	13	11	1	3
13	10	6	1	5	8	11	4	0	3	9	7	12	2
12	7	11	0	2	6	9	3	5	1	4	10	8	13
10	13	8	12	1	3	0	7	4	6	2	5	11	9
1	11	7	9	13	2	4	10	8	5	0	3	6	12
5	2	12	8	10	7	3	13	11	9	6	1	4	0

Proof. One checks directly that every row and every column of L_{14} is a permutation of $\{0, \dots, 13\}$, so L_{14} is a Latin square and $(\{0, \dots, 13\}, \cdot)$ with $a \cdot b = L_{14}[a][b]$ is a quasigroup. Computing $D_{14}[a][b] = L_{14}[b][L_{14}[a][b]]$ cell by cell yields the second table above (e.g. $D_{14}[0][2] = L_{14}[2][L_{14}[0][2]] = L_{14}[2][7] = 13$), and every row and column of D_{14} is likewise a permutation of $\{0, \dots, 13\}$. Hence the first recursive derivative is also a quasigroup, and L_{14} is recursively differentiable. Both Latin-square checks are finite and are reproduced by the verifier of Section 7. $\square$

Remark 5.2. L_{14} is noncommutative, has no two-sided identity (it is not a loop), and is nonassociative (not a group); its derivative differs from the base table. It is idempotent exactly on its first orbit (the diagonal reads $0, 1, \dots, 6$ on rows 0–6 and is fixed-point-free on rows 7–13), and is invariant under $g_{14} = (0 \cdots 6)(7 \cdots 13)$. Among the four readings of the “core” one might extract from a quasigroup — $b \cdot (a \cdot b)$, $a \cdot (a \cdot b)$, $(a \cdot b) \cdot b$, $(a \cdot b) \cdot a$ — only the correct one, $b \cdot (a \cdot b)$ of (2), produces a Latin derivative for this table; the object is keyed to the precise definition, not to a weaker variant.

6 A recursively differentiable quasigroup of order 18

Theorem 6.1. *The Cayley table L_{18} below is a quasigroup of order 18 whose first recursive derivative D_{18} is also a quasigroup. Hence a recursively differentiable quasigroup of order 18 exists.*

The base table L_{18} ($a, b = 0, \dots, 17$):

11	9	15	5	2	8	7	3	14	16	4	12	0	17	6	1	10	13
15	12	10	16	6	3	0	8	4	14	17	5	13	1	9	7	2	11
5	16	13	11	17	7	4	1	0	12	15	9	6	14	2	10	8	3
1	6	17	14	12	9	8	5	2	4	13	16	10	7	15	3	11	0
3	2	7	9	15	13	10	0	6	1	5	14	17	11	8	16	4	12
7	4	3	8	10	16	14	11	1	13	2	6	15	9	12	0	17	5
2	8	5	4	0	11	17	15	12	6	14	3	7	16	10	13	1	9
13	3	0	6	5	1	12	9	16	10	7	15	4	8	17	11	14	2
17	14	4	1	7	6	2	13	10	3	11	8	16	5	0	9	12	15
6	0	14	12	11	15	1	7	13	8	9	17	5	4	3	2	16	10
14	7	1	15	13	12	16	2	8	11	0	10	9	6	5	4	3	17
0	15	8	2	16	14	13	17	3	9	12	1	11	10	7	6	5	4
4	1	16	0	3	17	15	14	9	5	10	13	2	12	11	8	7	6
10	5	2	17	1	4	9	16	15	7	6	11	14	3	13	12	0	8
16	11	6	3	9	2	5	10	17	0	8	7	12	15	4	14	13	1
9	17	12	7	4	10	3	6	11	2	1	0	8	13	16	5	15	14
12	10	9	13	8	5	11	4	7	15	3	2	1	0	14	17	6	16
8	13	11	10	14	0	6	12	5	17	16	4	3	2	1	15	9	7

Its first recursive derivative $D_{18}[a][b] = L_{18}[b][L_{18}[a][b]]$:

12	14	10	9	7	1	15	6	0	16	13	11	4	8	5	17	3	2
1	13	15	11	10	8	2	16	7	3	17	14	12	5	0	6	9	4
8	2	14	16	12	11	0	3	17	5	4	9	15	13	6	1	7	10
9	0	3	15	17	13	12	1	4	11	6	5	10	16	14	7	2	8
5	10	1	4	16	9	14	13	2	0	12	7	6	11	17	15	8	3
3	6	11	2	5	17	10	15	14	4	1	13	8	7	12	9	16	0
15	4	7	12	3	6	9	11	16	1	5	2	14	0	8	13	10	17
17	16	5	8	13	4	7	10	12	9	2	6	3	15	1	0	14	11
13	9	17	6	0	14	5	8	11	12	10	3	7	4	16	2	1	15
7	15	2	10	14	0	8	9	5	13	11	4	17	1	3	12	6	16
6	8	16	3	11	15	1	0	10	17	14	12	5	9	2	4	13	7
11	7	0	17	4	12	16	2	1	8	9	15	13	6	10	3	5	14
2	12	8	1	9	5	13	17	3	15	0	10	16	14	7	11	4	6
4	3	13	0	2	10	6	14	9	7	16	1	11	17	15	8	12	5
10	5	4	14	1	3	11	7	15	6	8	17	2	12	9	16	0	13
16	11	6	5	15	2	4	12	8	14	7	0	9	3	13	10	17	1
0	17	12	7	6	16	3	5	13	2	15	8	1	10	4	14	11	9
14	1	9	13	8	7	17	4	6	10	3	16	0	2	11	5	15	12

Proof. As in Theorem 5.1: every row and column of L_{18} is a permutation of $\{0, \dots, 17\}$, so L_{18} is a quasigroup; the table $D_{18}[a][b] = L_{18}[b][L_{18}[a][b]]$ computed cell by cell is the second table, and every one of its rows and columns is a permutation of $\{0, \dots, 17\}$, so the derivative is a quasigroup. Hence L_{18} is recursively differentiable. The verifier of Section 7 reproduces both checks. $\square$

Remark 6.2. L_{18} is again idempotent-free, noncommutative, has no identity, and is nonassociative; it is invariant under $g_{18} = (0 \cdots 8)(9 \cdots 17)$, and only the reading $b \cdot (a \cdot b)$ of the core yields a Latin derivative.

This, with Theorem 5.1, proves Theorem 1.2 and hence Corollary 1.3.

7 Computational verification

The two theorems are finite statements verified directly from the definitions. The ancillary directory contains a standalone verifier, `rdiff_verify.py` (Python standard library only, no external data), which:

1. reads each printed table L_{14} , L_{18} (also stored as JSON);
2. checks that L is a Latin square (every row and column a permutation);
3. forms the derivative two independent ways — by the closed form $D[a][b] = L[b][L[a][b]]$ of (3), and by unrolling the recurrence (1) step by step — confirms the two agree, and checks that D is a Latin square;
4. runs the “wrong-reading” control: of the four candidate core operations only $b \cdot (a \cdot b)$ yields a Latin derivative, confirming the object is certified against the exact definition.

The verifier shares no code with the search that produced the tables, and a reader can reconstruct it in a few lines. The complete list of checks (each labelled to the claim it supports) and the search implementation are in the ancillary files; see `code/README.md`. The entire verification runs in well under a second.

8 Discussion

Corollary 1.3 closes the existence question of Conjecture 1.1 for binary quasigroups: a recursively differentiable quasigroup of order q exists if and only if $q \notin \{2, 6\}$. The two orders treated here were the last open cases on the loop-and-quasigroup problem list [4] (equivalently, the last two orders without a known full recursive $[4, 2, 3]_q$ code).

Two natural questions remain, neither needed for the conjecture but both of interest. First, the present objects are *ad hoc*: they are explicit witnesses found by search, not instances of a uniform family, and they do not *explain* existence at 14 and 18 the way a Mendelsohn-design construction explains it elsewhere. A uniform construction covering the orders $\equiv 2 \pmod 4$ that the PMD route skips would subsume the search. Second, both witnesses happen to carry a fixed-point-free automorphism with two regular orbits (a pair of $\frac{n}{2}$ -cycles with $\frac{n}{2}$ odd); whether *every* order $q \notin \{2, 6\}$ admits a recursively differentiable quasigroup with such symmetry, and more generally what the higher recursive-derivative degrees $v_r(2, q)$ are for 14 and 18 (we obtain only degree 1), is open. We hope the explicit tables are useful as test data for these questions.

Acknowledgements

This work was carried out under the Periapsis research structure with AI assistance (Claude). I thank the authors of [1, 2, 3] whose constructions and surveys framed the problem and pinned the two residual orders, and Parascovia Syrbu, who placed the small-order question on the open-problems list [4].

Honest status (read before relying on this). This is an unrefereed preprint, AI-assisted, with no journal referee check. Its central claims are finite and self-certifying: each Cayley table is printed in full and its recursive-differentiability is an $O(n^2)$ check reproduced by the standalone verifier, so the results are checkable without trusting the author or the assistant. The *novelty* claim — that no explicit order-14 or order-18 recursively differentiable quasigroup had previously been published — rests on a literature survey that reached the controlling sources ([3] first-hand, with its order- ≤ 100 table recording degree 0 at 14 and 18; the Moldovan-school survey tables recording “unknown” at both) but did *not* include a first-hand reading of the original 1998 source [1], which is in Russian and was consulted in translation/secondary form. A reader with access should confirm that point before treating the result as a priority claim rather than a clean construction.

References

- [1] E. Couselo, S. González, V. T. Markov, and A. A. Nechaev, *Recursive MDS-codes and recursively differentiable quasigroups*, Diskret. Mat. **10** (1998), no. 2, 3–29 (Russian); English transl. in Discrete Math. Appl. **8** (1998), no. 3, 217–246.
- [2] V. T. Markov, A. A. Nechaev, S. S. Skazhenik, and E. O. Tveritinov, *Pseudogeometries with clusters and a recursive $[4, 2, 3]_{42}$ -code*, J. Math. Sci. **163** (2009), 563–571 (English translation; the result settling $q = 42$ dates to 2008).
- [3] P. Klimov, *On the construction of recursively differentiable quasigroups and an example of a recursive $[4, 2, 3]_{26}$ -code*, arXiv:2604.01105 (2026).

- [4] M. Kinyon and P. Vojtěchovský (eds.), *Problems in loop theory and quasigroup theory*, open-problems list; problem “Small quasigroups with quasigroup core” (proposed by P. Syrbu, Loops '03, Prague, 2003). Maintained at https://en.wikipedia.org/wiki/Problems_in_loop_theory_and_quasigroup_theory (accessed June 2026).